

ALLIANCE KCSE 2025 PREDICTION



POSSIBLE KCSE EXAM



BIOLOGY

231/2

PAPER 2

(THEORY)

TIME: 2 HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

1. Write your name and Admission Number in the spaces provided above.
2. Sign and write the date of examination in the spaces provided above.
3. This paper consists of two sections: **A** and **B**. Answer **ALL** questions in section **A** in the spaces provided. In section **B**, answer question 6 (**compulsory**) and either question 7 or 8 in the spaces provided after question 8.

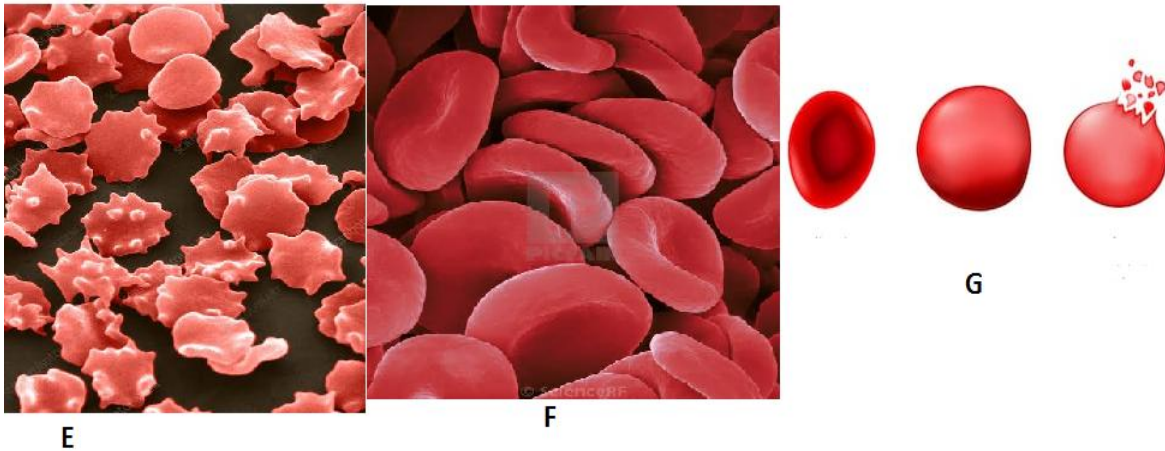
FOR EXAMINERS' USE ONLY

Section	Questions	Maximum Score	Candidate's Score
A	i)	8	
	ii)	8	
	iii)	8	
	iv)	8	
	v)	8	
B	vi)	20	
	vii)	20	
	viii)		
Total Score		80	

SECTION A (40 MARKS)

Answer ALL the questions in this section in the spaces provided.

1. The photograph below shows red blood cells that have been put in different solutions. Examine them and answer the questions that follow.



- a) i) Identify the type of solution in which F was placed.

(1mark)

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- ii) State the process which the red blood cells underwent in illustration G.

(1mark)

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- b) Account for the appearance the red blood cells underwent in illustration E.

(3marks)

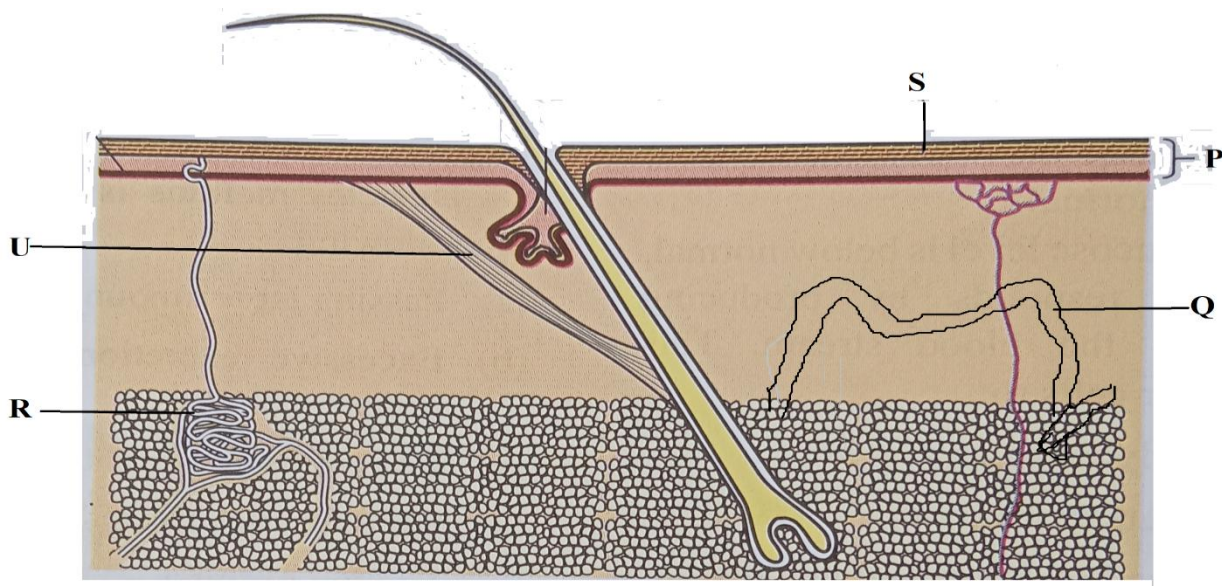
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- c) Explain what would happen if plant cells are placed in the solution in which the cells in G were immersed.

(3marks)

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2. The diagram below represents a vertical section through a mammalian skin.



- (a). Name the structures labelled P,Q,R and U

(4 marks)

P

Q

R

U

- (b) What is the function of S and T

(2marks)

S

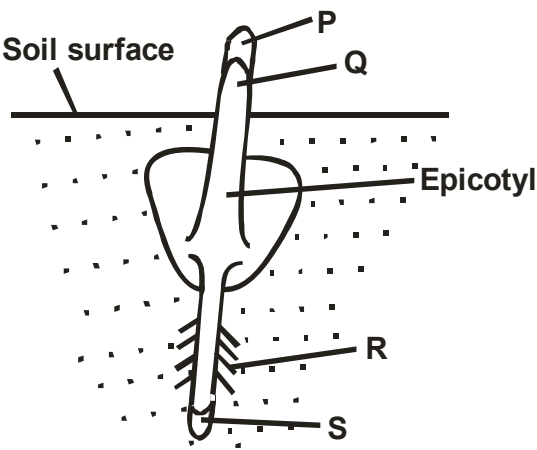
T

- (c) State the physiological changes that would occur in the following structures if the environmental temperature was raised gradually from 22°C to 40°C (2 marks)

R

Q.....

3. Diagram below represents a germinating seedling.



a) What is germination? (1 mark)

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b) Name the part labelled P, Q and R. (3 marks)

P

Q

R

c) Identify the type of germination shown in the diagram. (1 mark)

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What is the role of the following in germination of the above seedling?

i) Oxygen (1 mark)

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ii) Enzymes (1 mark)

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iii) Water (1 mark)

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4. The table below gives information about an aquarium which is ecologically balanced.

Type of organism	Dry weight
Insect	500gms
Fishes	1200gms
Water plants	5000gms
Bacteria	10gms

a) What do you understand by the term ecological balance? (1 mark)

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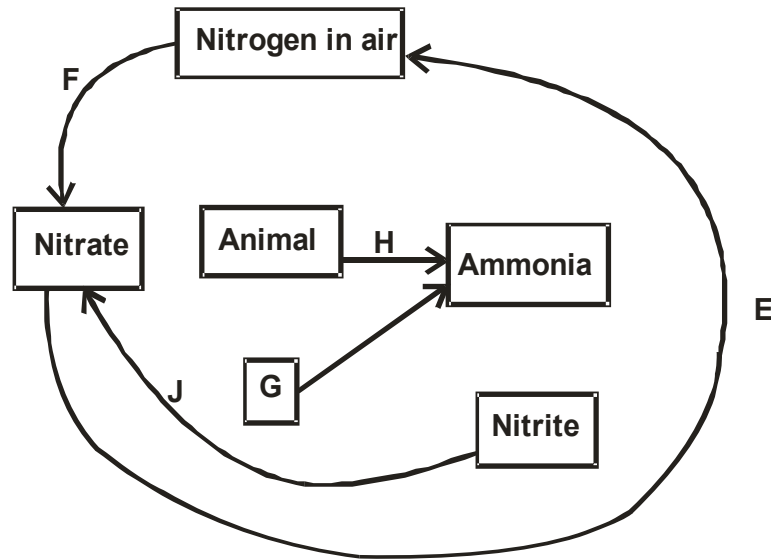
b) Calculate the total biomass of the aquarium. (Answer in kgs) (3 marks)

c) Which organism in the table is:

i) Primary producer.(1 mark)

ii) Secondary consumer.(1 mark)

d) The diagram below represents a simple nitrogen cycle.



i) Name the organisms that cause processes E and J. **(2 marks)**

E

J

5. When the offspring of purple and white flowered pea plants were crossed, they produced purple and white flowered plants in the ratio of 3: 1

Using letter H to represent the gene for purple colour

(a) State the genotype of:

(i) Parents **(2 marks)**

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(ii) F₁ Generation **(1 mark)**

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(b) Work out the cross between plants in the F₁ generation **(4 marks)**

(c) Account for the colour the flowers in plants of the F₁ generation (1mark)

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SECTION B

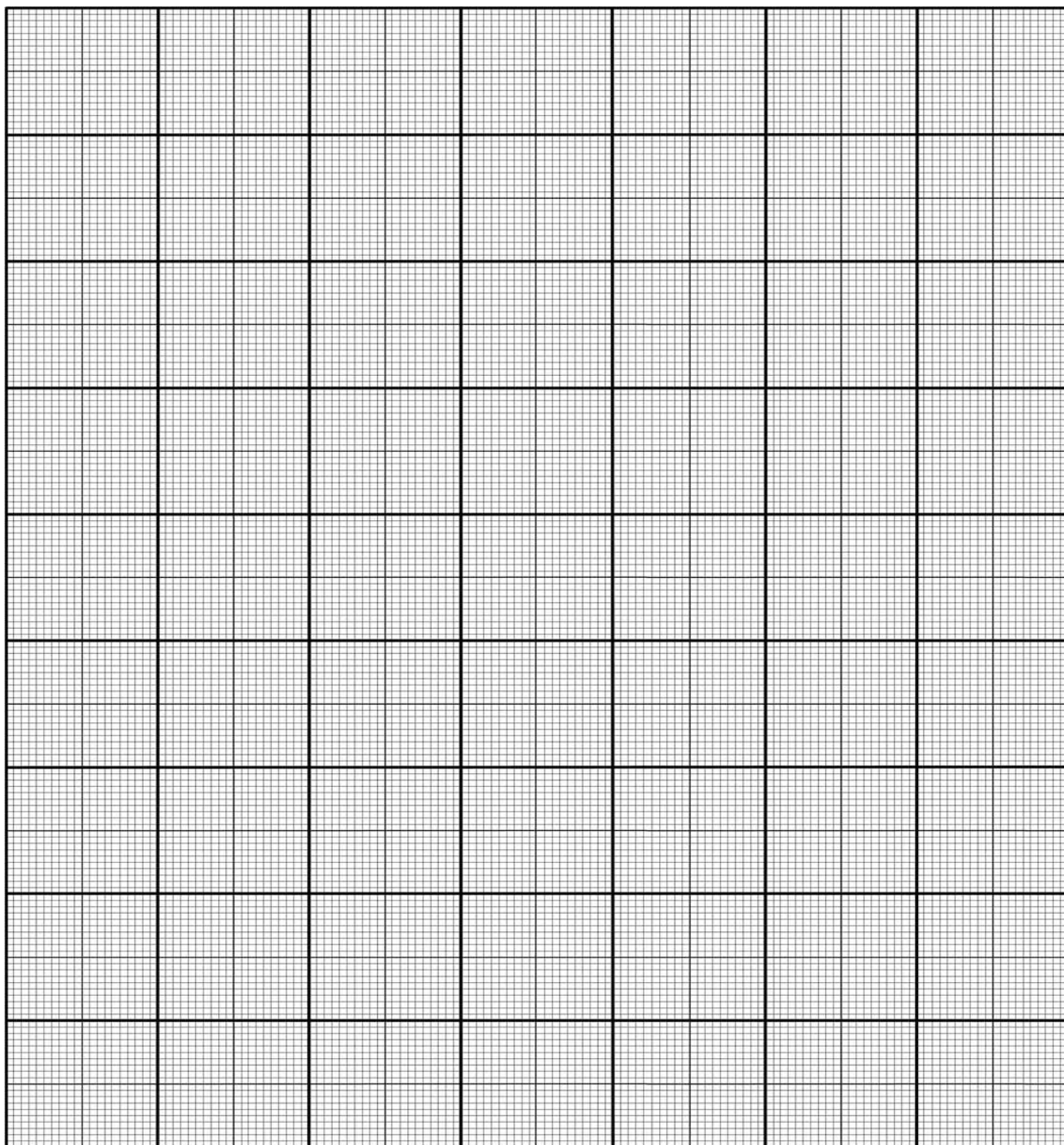
6. The data below represents levels of progesterone hormone produced in a female's body within a period of 34 days. Study the data and use it to answer the questions that follow

NB: The days were counted from the 1st day that menstruation was noticed.

Day	Progesterone hormone concentration in arbitrary units
1	6
2	5
3	3
4	2
5	1
6	1
8	1
10	2
12	4
14	7
16	8

20	9
22	10
24	10
26	10
28	10
30	11
32	11
34	11

(a) Plot a graph of progesterone concentration against time using a suitable scale. **(6marks)**



(b) Account for the progesterone levels in the blood between

4. Day 1 - day 5

(2marks)

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5. Day 14 – day 20 (2marks)

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6. Day 28 – day 35 (2marks)

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(c) Name two structures that produce progesterone in females . (3marks)

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(d) Suggest the process that usually takes place at day 14. (1mark)

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(e) Suggest two other hormones that were in high concentration in the body of the female between day 10 – 15 . Give reasons for your answer. (4marks)

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7. Describe the
- (i) Process of inhalation in mammals (10 marks)
 - (ii) Mechanism of opening and closing of stomata (10 marks)
8. How is the human eye adapted to its functions (20 marks)